

LAB WORKSHEET

Lab 8 — IP, VLAN, DNS, DHCP, FQDN, and Hosts File

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Objective: 2.2 — Network infrastructure services: IP, VLAN, DNS, FQDN, hosts file, RFC 1918, DHCP, APIPA.

Work through the lab on the Killercoda Ubuntu playground and record your results below.

Task 1 — Static + gateway

Assign a static IP and default gateway on eth0.

Paste `ip -c addr show eth0` and the default route.

Task 2 — VLAN tag

Create VLAN 20 on eth0 and address it.

Paste `ip -d link show eth0.20` confirming the vlan id.

Task 3 — DNS zone

Load the lab.local zone and resolve web.lab.local.

Record the resolved IP and the full FQDN.

Task 4 — DHCP scope

Write and validate the dhcpd.conf scope.

Paste the Config OK result and the pool range.

Task 5 — APIPA / hosts

Add an APIPA address and a hosts-file override.

Explain what a 169.254.x.x address indicates and why hosts beats DNS.

Completion checklist

- Configured a static IP, gateway and 802.1Q VLAN.
- Recited the three RFC 1918 ranges and APIPA.
- Resolved an A record from a BIND9 zone.
- Validated an ISC DHCP scope.
- Explained APIPA and the /etc/hosts override.